



GLOBALPAY DIGITAL FINANCIAL SERVICES

Enterprise Financial Technology Platform

Digital Wallets · Payment Processing · AI Fraud Detection · Open Banking · CBDC Simulation · Executive Reporting

Stage 1 — Enterprise Solution Presentation

Presenter

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- Diploma in Artificial Intelligence Operations
- EduQual Level 6

Project role & scope

- Solution provider for fictional client GlobalPay
- Modular monolith — 9 routers, 5 conceptual modules
- Simulated financial transactions only — no real banking

G Presentation Roadmap

Six sections — big picture first, then each module, then the data layer

1 Client & Problem

GlobalPay context, disconnected systems, transformation objective

2 FinTech Domain

Wallets, payments, Open Banking, CBDC and fraud analytics

3 Enterprise Architecture

How five modules communicate through secure APIs and data stores

4 Module Architectures

Wallet, payment, fraud AI, Open Banking/CBDC, executive dashboard

5 Network & Data Flows

TLS, REST APIs, PostgreSQL, DFD, ERD

6 Stack, Standards & Outcomes

Technology choices, framework mapping, limitations, future work

The complete set of 17 Section 8 deliverables is provided separately in the GitHub diagram pack

G Client Background, Problem & Objectives

Why GlobalPay needs this solution

Client context

- GlobalPay provides digital wallets, payments, merchant services, Open Banking APIs and analytics
- Fictional organisation for assessment purposes
- Processes 50M+ simulated transactions/day across countries

Business problem

- Disconnected wallet, payment, fraud and reporting systems
- Limited operational visibility across the business
- Fraud detection is reactive, not predictive
- Fragmented customer transaction analytics

Project objective

- Design a unified enterprise FinTech platform
- Demonstrate secure digital wallet and payment operations
- Add AI-assisted fraud detection with mandatory human review
- Simulate Open Banking and CBDC operations

Scope guardrails

- Simulated financial transactions only
- Laptop-scale proof of concept — 8-16 GB RAM
- Docker Compose deployment
- No real banking, card network or central bank connection

G

FinTech Domain Overview & User Roles

How wallets, payments, Open Banking, CBDC and fraud operations fit together

Role	What they see / do
Customer	Wallet, payments, profile, history
Merchant	Payments received, settlement view
Fraud analyst	Alert queue, review decisions
Analyst/Admin	API activity and health
Executive	Read-only KPIs and reports

Core concepts

- Digital wallet: simulated stored value, not a bank account
- Payment processing: validate, debit/credit, record atomically
- CBDC: simulated central-bank-issued digital currency
- Open Banking: consent-based third-party account/payment access
- AI fraud detection: risk-scored, human-reviewed, never auto-blocked

Management-facing (Modules 3-5)

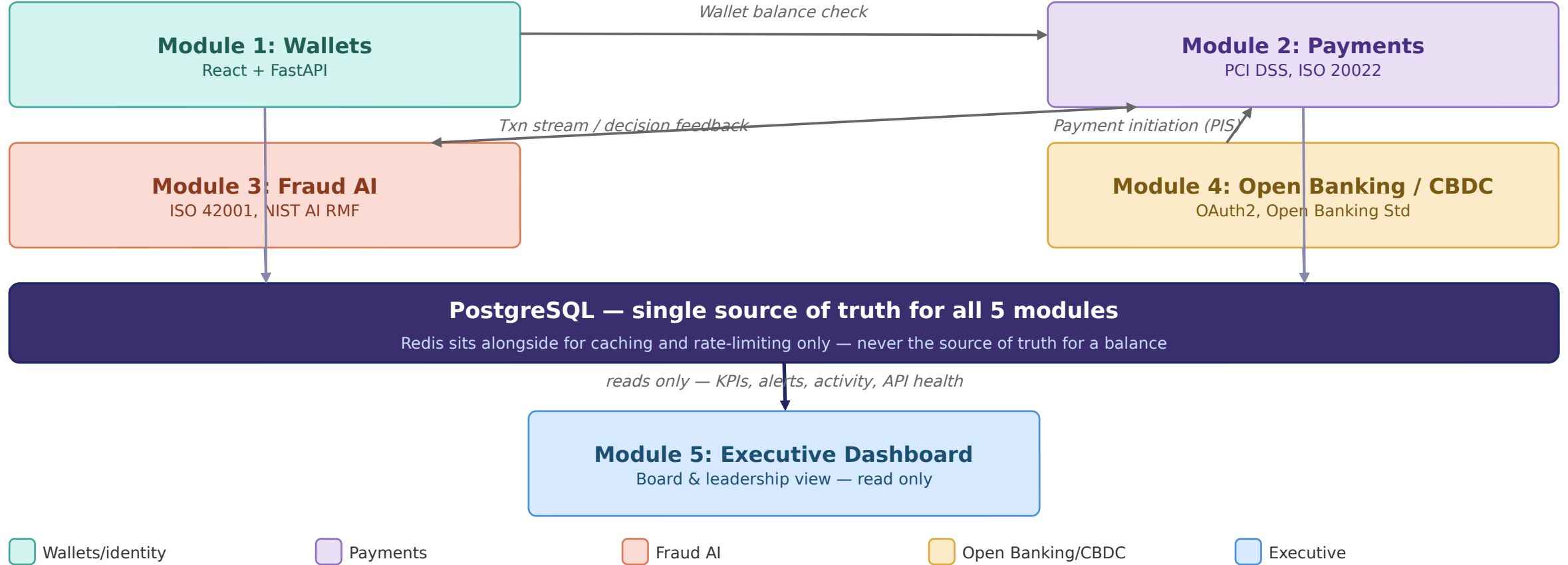
- Fraud alerts and investigation history
- API activity and health monitoring
- Executive KPIs and financial trend reports

Customer/merchant-facing (Modules 1-2)

- Registration, wallet balance and activity
- Payments, transfers and merchant charges
- Personal and settlement transaction history

G Enterprise Architecture

How the five modules connect — big picture before the detail



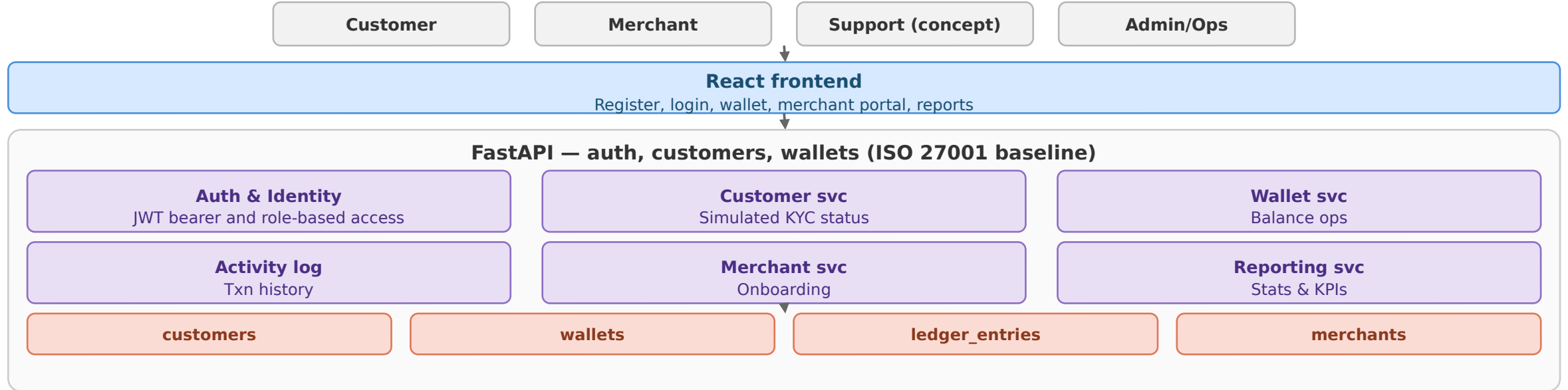
Rule: every module writes to PostgreSQL directly — the only module-to-module arrows are the two named above, and Module 5 reads, never writes



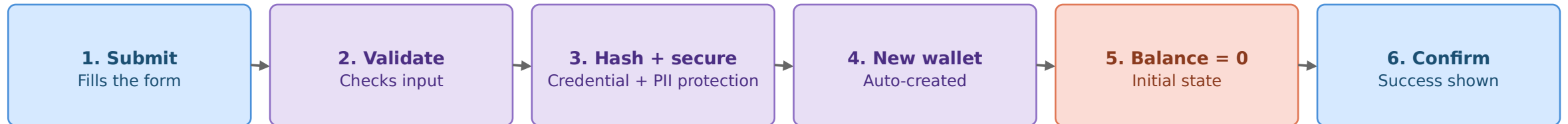
MODULE 1

Customer & Digital Wallet Management

Registration, wallet creation, balance operations, merchant accounts

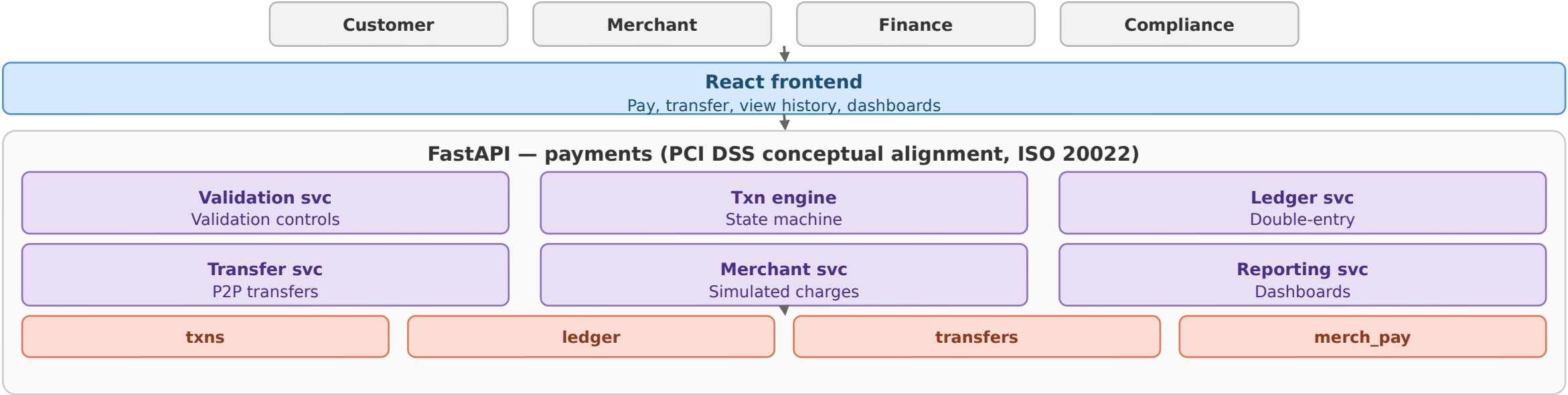


Registration & wallet creation:

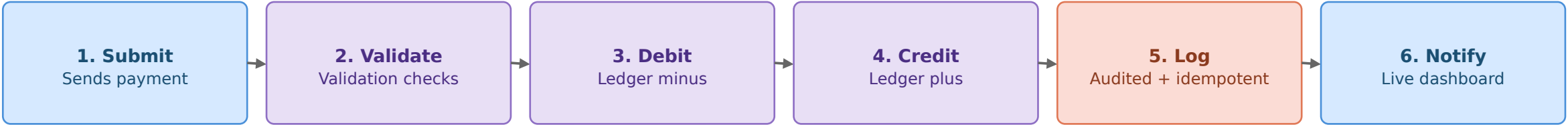


Argon2id password hashing + encrypted/masked PII fields — ISO 27018, COBIT 2019

Validation, double-entry ledger, transfers, merchant charges



Customer payment flow:

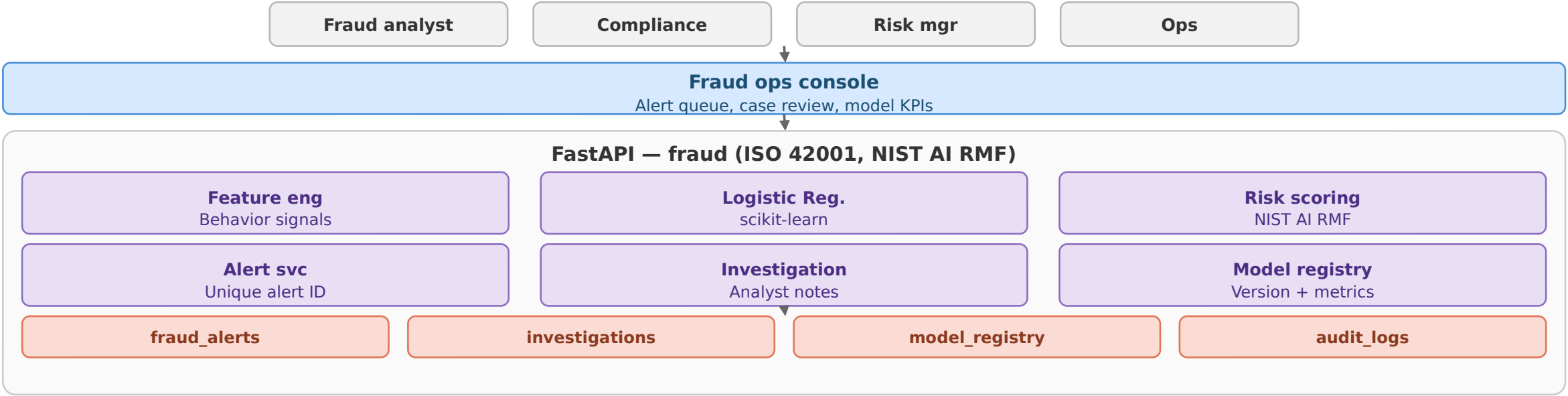


Two balanced debit/credit ledger rows are written atomically; idempotency blocks duplicate charges on retry

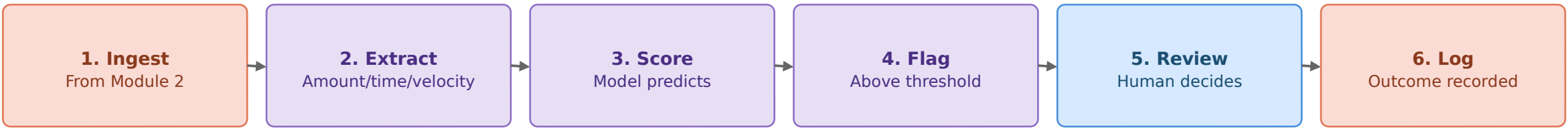


AI-Assisted Fraud Detection

Feature engineering, scoring, mandatory human review



Fraud review flow:

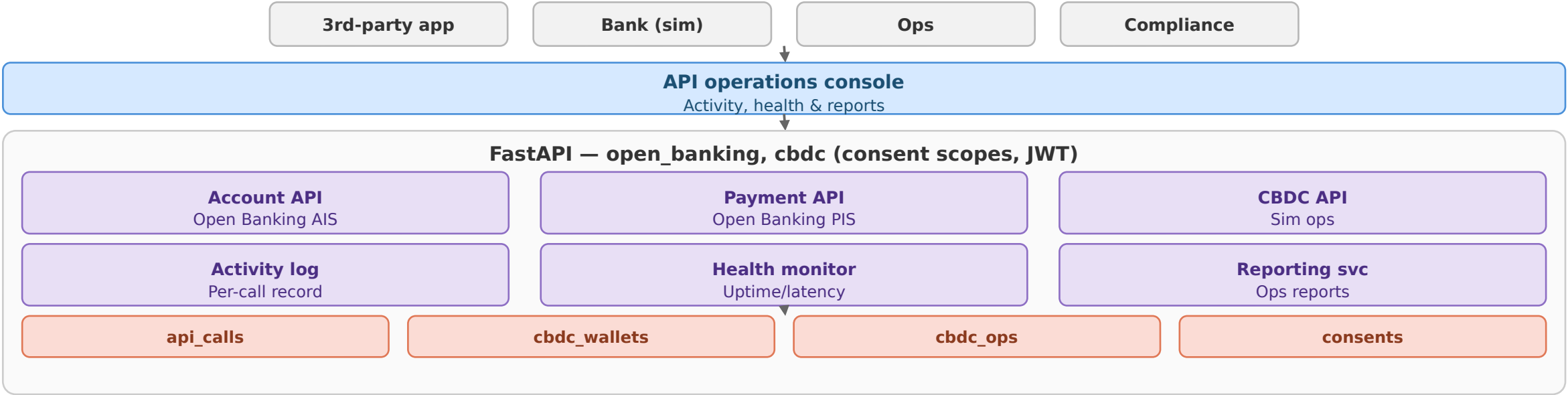


No automated blocking — human review is a project control aligned with ISO 42001 / NIST AI RMF

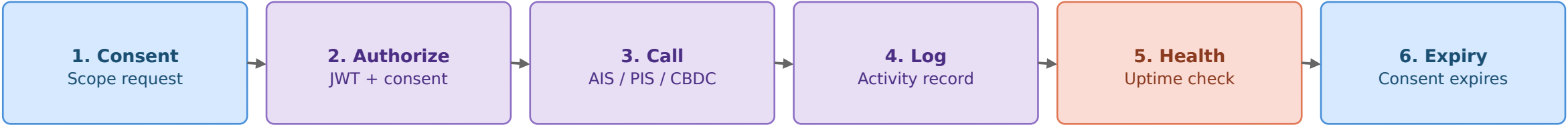


Open Banking & CBDC Simulation

Account/payment APIs, CBDC operations, activity monitoring



Open Banking / CBDC flow:



CBDC uses separate wallet and operation records from commercial wallets — no real central-bank connection

Executive Financial Operations Dashboard

Read-only aggregation of Modules 1-4 into leadership KPIs

CEO/Exec

Board/Auditor

Ops director

Compliance

Executive dashboard

KPIs, trends, alerts overview

FastAPI — dashboard (read-only, least privilege)

KPI aggregator

Read-only rollups

Trend engine

Time series

Alert summary

Cross-module

Client export

TXT/CSV/JSON

API/CBDC rollup

From Module 4

Access control

Read-only enforced

live KPI

trends

alert summary

report view

Executive reporting flow:

1. Pull data

Live query

2. Aggregate

Roll up KPIs

3. Trends

Time series

4. Flag

Cross-module

5. Report

Text export

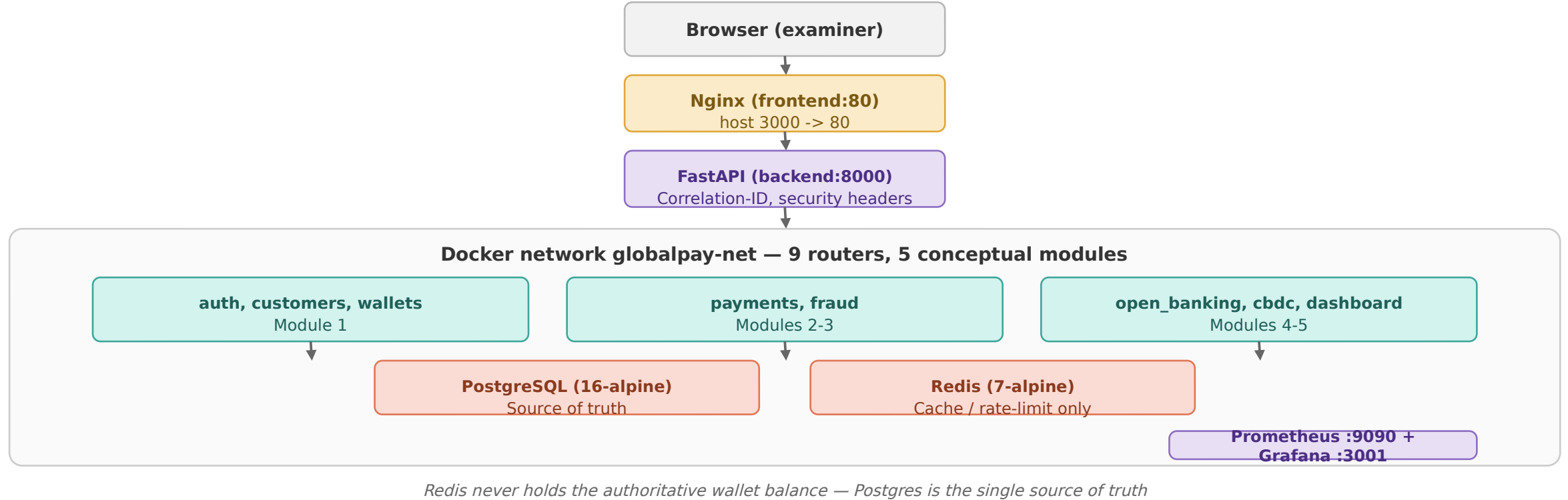
6. Present

Exec view

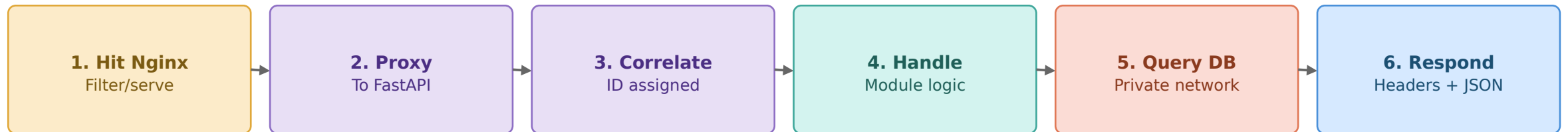
No write path back into any operational module — this is the least-privilege principle built into the architecture

G Enterprise Network Architecture

compose.yaml — 6 containers on one Docker network (globalpay-net)



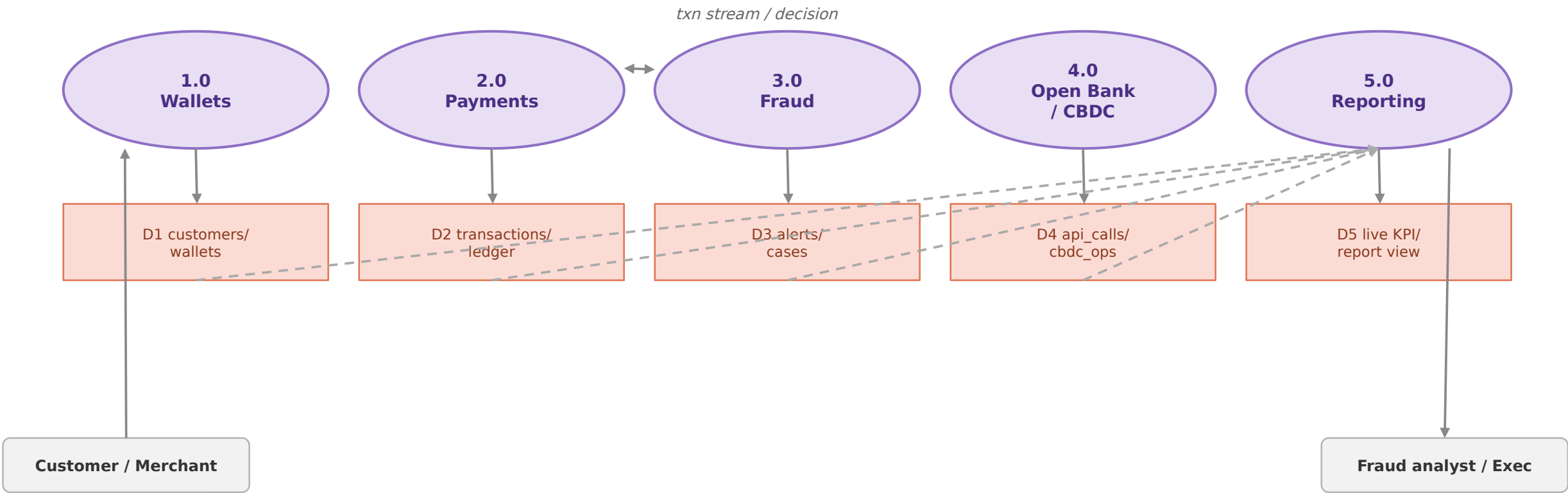
Request flow, step by step:





Data Flow Diagram

Level 0 context, decomposed into the five core processes at Level 1

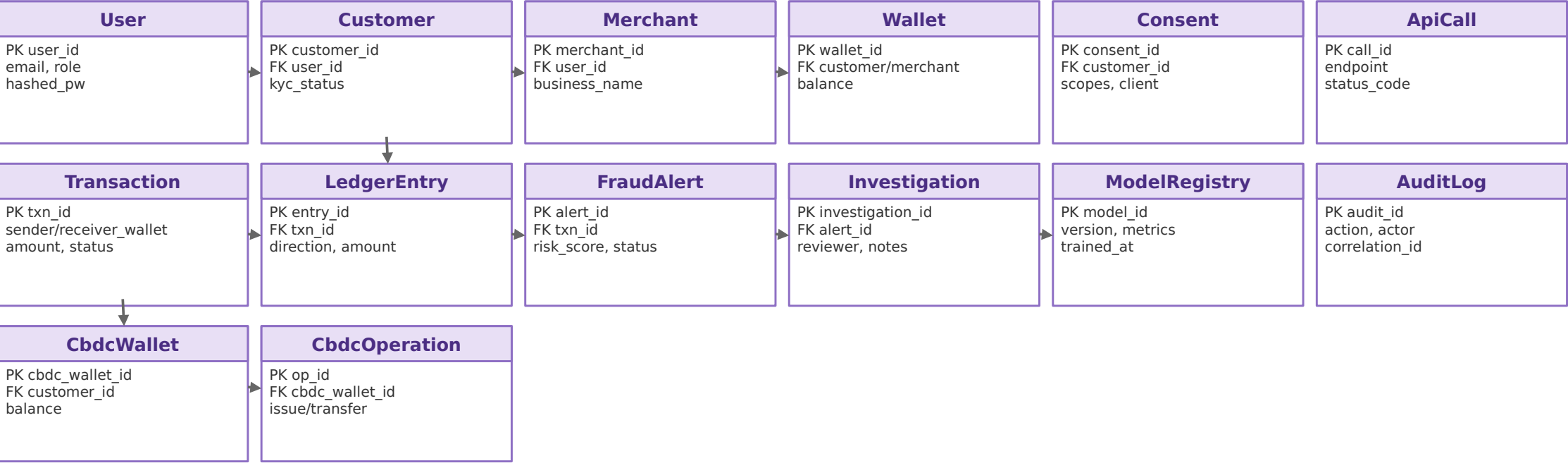


Level 0 (context view): the whole platform is a single process — "0 GlobalPay Platform" — with these same five entity groups around it. Level 1, shown above, decomposes that single process into its five internal parts.

Dashed arrows: on-demand aggregation from data stores into 5.0 Reporting

G Database Entity Relationship Diagram (ERD)

14 tables from backend/app/models/entities.py



User -> {Customer, Merchant}; Customer -> {Wallet, Consent, CbdcWallet}; Merchant -> Wallet. Transaction -> {LedgerEntry, FraudAlert -> Investigation}. ModelRegistry and AuditLog are cross-cutting.

Why this shape

- Foreign keys connect users, customers, wallets, transactions, alerts and investigations without orphaned records
- ModelRegistry and AuditLog sit outside the domain hierarchy on purpose — they observe every module, they belong to none
- CbdcWallet is a separate table from Wallet — central-bank money is never mixed with commercial wallet balances

G Technology Stack & Deployment Model

Laptop-optimised implementation, chosen for what's actually being assessed

Presentation

React web UI

API layer

FastAPI, OpenAPI/Swagger

AI / data

scikit-learn, Pandas, NumPy

Persistence

PostgreSQL + Redis

Docker Compose deployment boundary

Nginx

frontend

FastAPI

backend

PostgreSQL

database

Redis

cache

Prometheus

+ Grafana

Why Docker Compose, not K8s

- The module boundaries are what's assessed here, not container orchestration
- K3s/Kind remain available as an optional path
- Simple, repeatable setup for examiner demo

Security tooling

- Swagger + Postman/Newman API evidence
- Semgrep + Trivy scan; CIS Controls alignment
- Secrets kept outside source control

Deliberately excluded

- No real banking integration
- No locally hosted LLM, no GPU
- GitHub Actions CI; Kubernetes manifests optional

G Applicable Standards & Frameworks

Where each control applies across the platform

Framework	Applies to	Why it matters here
NIST SP 800-63	Module 1 — Customer svc	Conceptual identity guidance; no formal AAL claim
JWT bearer; OAuth2/OIDC future	Modules 1, 2, 4 — identity/API auth	JWT today; enterprise federation is future work
ISO/IEC 27018	Modules 1, 5 — PostgreSQL	Protection of customer PII
PCI DSS v4.0*	Module 2 — Validation svc	Conceptual alignment; no real card data processed
ISO 20022	Modules 2, 4 — messaging	ISO 20022-inspired fields; no full message schema
ISO/IEC 42001	Module 3 — Classifier	AI management system governance
NIST AI RMF	Module 3 — Risk scoring	Govern, Map, Measure, Manage
Open Banking Std	Module 4 — AIS/PIS API	Consent-based third-party access rules
ISO/IEC 27001	All modules — baseline	Overall information security management
ISO/IEC 27017*	All modules — cloud layer	Cloud shared responsibility, if cloud-deployed
OWASP Top 10 + API Top 10	All API layers	Access control, input validation, rate limiting, API risks
NIST CSF 2.0 + CIS Controls	Cybersecurity / deployment layer	Govern/Protect/Detect + secure configuration/logging
COBIT 2019	All modules — governance	Audit log, accountability, traceability



Business Outcomes, Limitations & Future Enhancements

How the solution improves operations today and how it could evolve

Expected business outcomes

- Unified view of wallets, payments, fraud and APIs
- Faster customer and merchant onboarding
- Clearer fraud investigation trail
- Executive visibility without operational access
- Audit-ready evidence across modules

Known limitations

- Simulated transactions only
- Laptop-scale proof of concept
- No real banks, card networks or central bank
- PCI DSS/ISO 27017 conceptual, not certified
- Single-node deployment, not high availability

Future enhancements

- Cloud deployment with private networking
- Enterprise IdP / OIDC federation
- Real Open Banking sandbox integration
- Event-driven architecture for scale
- Formal PCI DSS certification path



Thank You

Ready for questions — architecture, trade-offs, and live demonstration

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